PEER REVIEW HISTORY

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ARTICLE DETAILS

TITLE (PROVISIONAL)	Effects of the Maternal and Child Health handbook and other home-based records on mothers' non-health outcomes: a systematic review
AUTHORS	Carandang , Rogie Royce; Sakamoto, Jennifer Lisa; Kunieda, Mika; Shibanuma, Akira; Yarotskaya, Ekaterina; Basargina, Milana; Jimba, Masamine

VERSION 1 – REVIEW

REVIEWER	Nasir, Narila Mutia Universitas Islam Negeri Syarif Hidayatullah Jakarta
REVIEW RETURNED	16-Dec-2021

GENERAL COMMENTS	#1 In the abstract part, it is suggested to write down what outcome that categorized as non-health outcomes #2 In the Introduction part, please emphasize why knowing the non-health outcomes is important, for example it will impact to health outcome as well #3 In the methods part, please clearly defined the code "mixed" for outcome #4 In the result part line 291, please write down how many studies that not using comparison group #5 In the discussion part, please give more explain on the issue new finding such as why it is found specifically in Japan, for example maybe because the using of MCH in Japan has been for many years? (line 418) On the issue of inconsistency of Health Provider, Please add more explanation if there is any reference to support the statement. Line 476, the new finding and need special attention, could you explain what kind of not relevant information giving by health provider? #6 In the conclusion part, the health provider not only need training but also how to monitor and evaluate what they are doing in the utilization of home-based record is also important #7 the terms "health setting" refers to health facility setting? If Yes, it is better to use the health facility setting

REVIEWER	Nicol, Jeannine Stellenbosch University, Global Health
REVIEW RETURNED	18-Jan-2022

GENERAL COMMENTS	In general, the paper is well written and focuses on an important subject. However, the paper could be improved by providing clarity on the following issues & concerns: 1) The aim(s) of the paper presented in the abstract seems to be different from what is listed in the introduction section. There should be a level of coherence between the aims written in the abstract with the ones in the main text 2) It's not clear whether the focus on the review is on Japanese population or across the world. some section of the background focuses on japan but the included papers seems to from both high and low -middle income countries. the authors should clarify 3) the outcomes measures are not well defined particularly the primary outcomes. This will be useful to guide the reader on the inclusion criteria of the studies 4) There is no clear definition how the measure of effect has been reported /computed . given the focus of the review, this is critical and clarity is needed 5) The heterogeneity of the included paper could be explained by the broad spectrum of the types of study design included in the review. this should be reviewed in light with the focus of the aims of the review - measure the effect of Road to health card

REVIEWER	gholipour, kamal Tabriz University of Medical Sciences, Health services management
REVIEW RETURNED	22-Jan-2022

GENERAL COMMENTS	This paper focused on an important subject. Please address the following points before more consideration: • Your main outcome in SR registration form in PROSPERO were health outcome and quality of care for Newborn/child, so in this manuscript you focused mainly on non-health aspects of mothers' care. It appears you needs to a revise your search strategy and search terms to cover all relevant studies. Which one of your search strategies refers to mothers? • There are many relevant studies using home medical records to participate women in their own maternity services: a. Gholipour K, Tabrizi JS, Jafarabadi MA, Iezadi S, Mardi A. Effects of customer self-audit on the quality of maternity care in Tabriz: A cluster-randomized controlled trial. PloS one. 2018 Oct 11;13(10):e0203255. b. Gholipour K, Tabrizi JS, Asghari Jafarabadi M, Iezadi S, Farshbaf N, Farzam Rahbar F, Afsharniya F. Customer's self-audit to improve the technical quality of maternity care in Tabriz: a community trial. EMHJ-Eastern Mediterranean Health Journal. 2016;22(5):309-17.) • There is an updated PRISMA 2020 which should be used rather than the 2009 version. • In Table S2 you presented study setting ambiguously; in some cases refer to country or geographical area and in some cases the health care settings.

VERSION 1 – AUTHOR RESPONSE

Response to Reviewer 1 Comments

Point 1: In the abstract part, it is suggested to write down what outcome that categorized as non-health outcomes.

Response 1: Thank you very much for your suggestion. We have revised the abstract as per editor's suggestion and adhered to the PRISMA extension for abstracts to ensure that all relevant information was within the abstract.

Point 2: In the Introduction part, please emphasize why knowing the non-health outcomes is important, for example it will impact to health outcome as well.

Response 2: Thank you very much for your comment. We have emphasized why non-health outcomes is important. Please refer to page 4, lines 123-128.

Exploring these non-health outcomes can be crucial for providing a more holistic picture of the effectiveness of home-based records and result in insights of theoretical and practical relevance.[14-17] This would capture the user experience to help improve the implementation of home-based records. Moreover, non-health outcomes may impact health outcomes,[14] although more studies need to be conducted to clarify this effect.

Point 3: In the methods part, please clearly defined the code "mixed" for outcome.

Response 3: Thank you very much for your critical comment. We have explained the coding process of the outcomes in more detail. Please refer to pages 8-9, lines 263-273.

To evaluate the effects of the intervention (home-based records), we conducted a detailed examination of the numeric and textual summary of the findings and conclusions of the included studies. We coded the outcomes as having a positive, mixed, or no effect. We considered an outcome to have a 'positive effect' if the home-based record showed a statistically significant effect (e.g., women experienced more partner involvement) and narrative findings indicated positive results (e.g., healthcare providers explained what is being recorded). We coded an outcome to have a 'mixed effect' when it showed some evidence of the usefulness of the record but not necessarily a significant effect. When there was no significant effect and narrative findings reported negative results (e.g., perceived lack of communication with healthcare providers), we considered the outcome as 'no effect.'

Point 4: In the result part line 291, please write down how many studies that not using comparison group.

Response 4: We have added how many studies without a comparison group in the results section. Please refer to page 10, lines 314-316.

Some studies did not include a comparison group (n = 33) when evaluating the intervention, while others compared users of home-based records with non-users of records or standard care groups.

Point 5: In the discussion part, please give more explain on the issue new finding such as why it is found specifically in Japan, for example maybe because the using of MCH in Japan has been for many years? (line 418)

Response 5: We have presented the mechanisms of how mother-child bonding was seen among the MCH handbook users in Japan. This is related to the integration of MCH records and how mothers in Japan use the handbook. Please refer to page 19, lines 441-463.

Of the different types of home-based records, only the MCH handbook may have fostered mother-child bonding. This finding is new and is only found in Japanese articles. Various ways could explain how the use of the MCH handbook facilitated mother-child bonding. First, the handbook was considered a special gift, filled with parental love and mothers' messages for their children, given to children during their marriage or pregnancy.[74-75] Mothers in Japan wrote down their worries, joy, and expectations from pregnancy and child rearing in the handbook, along with some healthy behaviors that could be passed on to the next generation.[75, 80] Losing these handbooks to a natural disaster was a painful experience for Japanese mothers, as it meant losing all their pregnancy and child health records.[79] Second, the handbook could be used to predict the child's school adaptation,[78] and the possible course of pregnancy and delivery for the daughter.[76] That is, school maladaptation was evident among children whose mothers had recorded at least one

incident of worry or anxiety in the MCH handbook. This can be attributed to the fact that the emotional bond with the mother is critical for the child's social, emotional, and cognitive development.[81-83] Thus, the mother's worry or anxiety is likely to hinder the development of such a bond, leading to difficulties in adaptation for the child. Third, it served as an aide-memoire for mothers who had experienced neonatal death.[35-36] Mothers' words of gratitude written in the handbook served as evidence of the bonds formed during pregnancy. Finally, the handbook served as a tool to help reduce parental anxiety and build good parent-child relationships, even among adoptive parents.[77] Overall, the findings showed that the MCH handbook is an essential source of information to learn more about the mother-child relationship. The bonding formation may be attributed to the integration of MCH records and how mothers in Japan use the handbook.

Point 6: On the issue of inconsistency of Health Provider, please add more explanation if there is any reference to support the statement.

Response 6: We added a new reference to support the inconsistency of reporting among healthcare providers. Please refer to page 20, lines 486-495.

However, we observed inconsistencies in the use of records across health facilities and professionals, which might discourage mothers from using home-based records. Private clinics and hospitals were less likely to use the records than public and primary care settings.[43, 46, 67] Moreover, doctors (e.g., general practitioners, pediatricians) were less likely to use and refer to home-based records than nurses and health visitors during check-ups and consultations.[57, 67, 69]; this finding is consistent that from a previous systematic review.[11] Generally, community nurses are the most likely professionals to use/refer to the home-based records in the health facilities.[11] Such reluctance from doctors to fill out a home-based record may arise if they are not properly oriented to see the benefits of using these records for themselves and their patients.

Point 7: Line 476, the new finding and need special attention, could you explain what kind of not relevant information giving by health provider?

Response 7: Thank you very much for your clarification. We have added more information about the ethnic minority women in Vietnam. Please refer to pages 20-21, lines 502-510.

Furthermore, ethnic minority women in Vietnam reported receiving health information from providers that was non-specific and not relevant to their context.[34] For instance, they were not given specific dietary advice and told to eat from all food groups and take iron supplements when they 'lack blood,' which is unclear how they would assess this themselves.[34] This finding is new and requires special attention. That is, it is imperative that the handbook offers personalized guidance, especially for women with lower education and from minority populations. This can help build trust and strong partnerships between mothers and healthcare providers and reduce barriers for women in accessing healthcare [38, 84-85].

Point 8: In the conclusion part, the health provider not only need training but also how to monitor and evaluate what they are doing in the utilization of home-based record is also important.

Response 8: Thank you very much for raising this important point. We have added your suggestion in the conclusion. Please refer to page 22, lines 547-551.

Thus, various types of training must be conducted at the local level across health facilities and for all healthcare professionals to orient them to the use and benefits of home-based records and, therefore, help them provide patient-centered care. Moreover, we should monitor and evaluate the use of the MCH handbook and other home-based records to ensure their effective implementation.

Point 9: The terms "health setting" refers to health facility setting? If Yes, it is better to use the health facility setting.

Response 9: Thank you very much for your comment. We used the term 'health facilities' throughout the text.

Response to Reviewer 2 Comments

Point 1: In general, the paper is well written and focuses on an important subject. However, the paper could be improved by providing clarity on the following issues & concerns:

Response 1: Thank you very much for reviewing our manuscript. We are glad to hear about your comments and appreciation of our study.

Point 2: The aim(s) of the paper presented in the abstract seems to be different from what is listed in the introduction section. There should be a level of coherence between the aims written in the abstract with the ones in the main text.

Response 2: Thank you very much for pointing this out. We updated the objective in the abstract and main text. Please refer to the abstract and the main text on page 5, lines 137-139.

Objective This review aimed to investigate the effects of the maternal and child health (MCH) handbook and other home-based records on mothers' non-health outcomes.

In light of these gaps left unaddressed by existing literature, the present study aimed to investigate the effects of the MCH handbook and other home-based records on mothers' non-health outcomes, through a review of studies published in English and Japanese.

Point 3: It's not clear whether the focus on the review is on Japanese population or across the world. Some section of the background focuses on Japan, but the included papers seem to from both high and low -middle income countries. The authors should clarify.

Response 3: Thank you very much for your clarification. The MCH handbook is not only used in Japan. To date, more than 50 countries worldwide have used the handbook, and it was found to be useful in low- and middle-income countries. Please refer to page 4, lines 105-107.To date, more than 50 countries worldwide have used the MCH handbook and found it to be useful.[4] This is especially true for countries where access to healthcare services is restricted.[9]

We have presented a brief history of the MCH handbook as this will interest the readers in how it was created and implemented. Nevertheless, we added more information about home-based records used in other countries to address your concern. Please refer to page 3, lines 76-91.

Over 163 countries worldwide have made use of home-based records to improve maternal, newborn, and child health (MNCH).[1] Home-based records are handheld records used by mothers or caregivers in households to record essential information related to MNCH, including visits to a healthcare provider, vaccination history, and the child's developmental milestones.[1] The design and content of these records vary considerably across countries and regions. While their use is nearly universal in some countries, it tends to be limited in others.[1] The records are available in paper or electronic format, complement facility-based records, and can be either single- or multi-focus. Single-focus records contain information relevant to one health topic or population group (e.g., antenatal care notes, vaccination-only cards, growth charts), while multi-focus records consist of chronologically ordered information pertaining to more than one health topic and can be used for an extended period.[2] The difference in focus as per health topic or population group resorted to policy debates on whether home-based records should be developed and distributed per mother or child.[3] Due to problems encountered in full integration (e.g., poor coordination across stakeholders), most countries prefer to implement program-specific, stand-alone home-based records for MCH services.[3]

Point 4: The outcomes measures are not well defined particularly the primary outcomes.

This will be useful to guide the reader on the inclusion criteria of the studies.

Response 4: Thank you very much for your constructive comment. We have defined the outcome measures in detail. Please refer to page 6, lines 172-185.

Outcome. We followed the WHO guidelines for defining non-health outcomes.[1] These included communication within the household, communication with healthcare providers, satisfaction with home-based records, and satisfaction with services/provider performance.[1] Communication within the household refers to how home-based records improved partner/family members' involvement in pregnancy and childcare, while communication with healthcare providers covers counseling sessions using the records and mothers' engagement. Satisfaction with home-based records refers to mothers' perceived agreement with its content (e.g., health or recording information). In contrast, satisfaction with services/provider performance refers to mothers' perceived use of the records to deliver MCH services. As an additional outcome, we included mother-child bonding based on the assumption that the integration of the mother's and child's records in the MCH handbook can foster a stronger mother-child bond. We defined 'mother-child bonding' as the development of a core relationship between mother and child.[20] This bond is unidirectional (from mother to child), shapes during pregnancy, and continues developing until early childhood.[21-23]

Point 5: There is no clear definition how the measure of effect has been reported /computed. Given the focus of the review, this is critical and clarity is needed

Response 5: Reviewer 1 also raised the same concern. We have described the measure of effect in detail. Please refer to response 3 in Reviewer 1.

Point 6: The heterogeneity of the included paper could be explained by the broad spectrum of the types of study design included in the review. This should be reviewed in light with the focus of the aims of the review - measure the effect of Road to health card.

Response 6: We agree with you. We have written this in our limitations section. Please refer to page 21, lines 531-534.

Third, we observed marked heterogeneity across studies regarding the study designs, intervention types, and comparator groups, all of which may have modified the study outcomes. Hence, we conducted a narrative synthesis, and evaluated the risk of bias and certainty of evidence for all included studies.

Response to Reviewer 3 Comments

Point 1: This paper focused on an important subject. Please address the following points before more consideration:

Response 1: Thank you very much for reviewing our manuscript. We are glad to hear about your comments and appreciation of our study.

Point 2: Your main outcome in SR registration form in PROSPERO were health outcome and quality of care for Newborn/child, so in this manuscript you focused mainly on non-health aspects of mothers' care. It appears you needs to a revise your search strategy and search terms to cover all relevant studies. Which one of your search strategies refers to mothers?

Response 2: Thank you very much for your constructive comment. We have updated the search strategy and did the database search again to cover all relevant studies. Please refer to Text S2 for the search strategy and Figure 1 for the updated PRISMA 2020 flowchart.

Point 3: There are many relevant studies using home medical records to participate women in their own maternity services:

a. Gholipour K, Tabrizi JS, Jafarabadi MA, Iezadi S, Mardi A. Effects of customer self-audit on the quality of maternity care in Tabriz: A cluster-randomized controlled trial. PloS one. 2018 Oct 11;13(10):e0203255.

b. Gholipour K, Tabrizi JS, Asghari Jafarabadi M, Iezadi S, Farshbaf N, Farzam Rahbar F, Afsharniya F. Customer's self-audit to improve the technical quality of maternity care

in Tabriz: a community trial. EMHJ-Eastern Mediterranean Health Journal. 2016;22(5):309-17.)

Response 3: Thank you very much for your suggestion. We added the PLOS ONE article to our systematic review. The second article is similar to the first, so we did not include it.

Point 4: There is an updated PRISMA 2020 which should be used rather than the 2009 version.

Response 4: As you and the editor suggested, we used PRISMA 2020 throughout the manuscript.

Point 5: In Table S2 you presented study setting ambiguously; in some cases, refer to country or geographical area and in some cases the health care settings.

Response 5: Thank you very much for your comment. We want to report the study setting as specific as possible. We are only limited to the information provided in each study included in this review.

VERSION 2 – REVIEW

REVIEWER	Nasir, Narila Mutia Universitas Islam Negeri Syarif Hidayatullah Jakarta
REVIEW RETURNED	24-Apr-2022

GENERAL COMMENTS	Thank you very much for your response. It becomes more clear now. The paper is well written and discussing an interesting topic. I really appreciate the authors' effort. Just for suggestion, if your study is the first study which combine the review paper both in English and Japanese, maybe you can emphasize in the introduction section to indicate the novelty of your study. Thank you.

REVIEWER	gholipour, kamal Tabriz University of Medical Sciences, Health services management
REVIEW RETURNED	23-Apr-2022

GENERAL COMMENTS	No further comments.